

Functional Annotation Report: *lgt* (Q88CN8, PP_5142) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

- **UniProt:** Q88CN8
- **Gene:** *lgt* (OrderedLocusName PP_5142)
- **Protein:** Phosphatidylglycerol–prolipoprotein diacylglyceryl transferase; EC 2.5.1.145
- **Organism:** *Pseudomonas putida* KT2440 (PSEPK), a Gammaproteobacterium
- **Family/Domains:** Lgt family (IPR001640; Pfam PF01790); UniProt annotation derived from HAMAP rule MF_01147

Identity confirmed — no gene-symbol ambiguity. The gene symbol *lgt*, the EC number, the "phosphatidylglycerol–prolipoprotein diacylglyceryl transferase" description, and the Lgt/PF01790 family assignment all correspond precisely to a single, well-characterized and highly conserved bacterial enzyme. All literature retrieved (*E. coli*, *Corynebacterium*, *Listeria*, and pan-bacterial evolutionary/structural studies) describes exactly this enzyme. Because *P. putida* KT2440 itself has not been the subject of a dedicated Lgt biochemical study, function below is assigned from (i) the highly conserved enzyme family and HAMAP rule, (ii) mechanistic/structural work in model bacteria (directly transferable given the family's strong conservation), and (iii) direct sequence analysis of the Q88CN8 protein performed during this investigation.

Summary

The gene *lgt* (UniProt Q88CN8; ordered locus PP_5142) of *Pseudomonas putida* KT2440 encodes **phosphatidylglycerol–prolipoprotein diacylglyceryl transferase (Lgt; EC 2.5.1.145)**, an integral inner-membrane enzyme that catalyzes the **first and committed step of bacterial lipoprotein biogenesis**. In this reaction, Lgt transfers the *sn*-1,2-diacylglyceryl moiety from

the membrane phospholipid **phosphatidylglycerol** onto the sulfhydryl (thiol) side chain of the **invariant cysteine** within the "lipobox" motif of prolipoprotein signal sequences, generating a thioether-linked *S*-diacylglyceryl-prolipoprotein and releasing *sn*-glycerol-1-phosphate. This modification tethers otherwise-soluble secreted proteins to the membrane and licenses the downstream, universally conserved maturation steps (**Lgt** → **LspA** → **Lnt**).

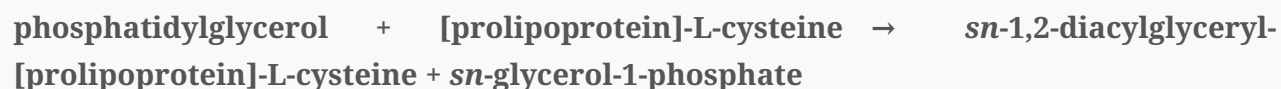
The identity of the *P. putida* protein is well supported. The gene symbol, EC number, Lgt family assignment (**IPR001640** / **Pfam PF01790**), and HAMAP rule MF_01147 all agree with the biochemically characterized enzyme family. Direct sequence analysis performed during this investigation confirmed that the 268-residue *P. putida* ortholog **retains the two catalytic arginines** shown experimentally to be essential in *Escherichia coli* Lgt (Arg143 and Arg239), and hydropathy analysis confirmed the expected **polytopic (~7 transmembrane helix) inner-membrane architecture**. Lgt *modifies* other proteins for membrane anchoring but is **not itself a lipoprotein** — it has no cleavable signal peptide or lipobox of its own.

Functionally, Lgt is **essential for viability in Proteobacteria**, the clade that contains *P. putida*. Its loss permeabilizes the outer membrane, causes abnormal morphology and cell death, and sensitizes cells to serum and antibiotics. Because it is essential, membrane-embedded, and structurally conserved, Lgt is an actively pursued **antibacterial drug target**. The enzyme performs its function at/within the **cytoplasmic (inner) membrane**, using two lipid-binding sites and a mechanism in which both the phospholipid substrate and the lipid-modified peptide product enter and exit the active site **laterally** through the lipid bilayer.

Key Findings

Finding 1 — Lgt catalyzes the first, committed step of lipoprotein biogenesis

Lgt initiates lipoprotein maturation. The reaction it catalyzes (EC 2.5.1.145) is:



- **Lipid donor:** phosphatidylglycerol (PG), a native inner-membrane phospholipid.
- **Protein acceptor / substrate specificity:** prolipoproteins bearing a **lipobox** motif (consensus [LVI]-[ASTVI]-[GAS]-C) at the end of their signal sequence. The invariant cysteine

is the sole acceptor, and the diacylglyceryl group is attached via a thioether bond to its sulfhydryl.

Substrate recognition and chemistry were defined structurally by Noland and colleagues, who describe that "*phosphatidylglycerol:prolipoprotein diacylglyceryl transferase (Lgt) recognizes a conserved lipobox motif within the prolipoprotein signal sequence and catalyzes the addition of diacylglycerol to an invariant cysteine*" (PMID: 28698362). This is the **first and committed step** of the pathway — once the diacylglyceryl group is attached, the substrate is irreversibly channeled toward becoming a mature lipoprotein. Legood and colleagues state that "*prolipoprotein phosphatidylglycerol diacylglyceryl transferase (Lgt) catalyzes the first and committed step*" of lipoprotein modification (PMID: 42126242). This designation is functionally important: Lgt sets the flux through the entire downstream maturation and sorting machinery.

The functional assignment for Q88CN8 derives from the Lgt family assignment (IPR001640, Pfam PF01790) and HAMAP rule MF_01147, both of which map the protein onto this biochemically characterized enzyme class.

Finding 2 — Lgt is an integral membrane enzyme with two lipid-binding sites and a lateral catalytic mechanism

The three-dimensional structure of *E. coli* Lgt was solved by Mao and colleagues in complex with its substrate **phosphatidylglycerol** (1.9 Å) and with the inhibitor **palmitic acid** (1.6 Å). These structures revealed **two distinct lipid-binding sites** and identified the catalytic architecture. The authors confirm that "*Lgt is an integral membrane enzyme that catalyses the first reaction of the three-step post-translational lipid modification*" (PMID: 26729647).

Two conserved arginine residues, **Arg143 and Arg239**, were shown by complementation of *lgt*-knockout cells to be **essential for diacylglyceryl transfer**: "*revealed critical residues, including Arg143 and Arg239, that are essential for diacylglyceryl transfer*" (PMID: 26729647). These arginines are thought to stabilize the phosphate of the PG substrate / transition state and position it for nucleophilic attack by the cysteine thiolate.

A defining mechanistic feature is that both the phospholipid substrate and the lipid-modified peptide product move **laterally** in and out of the membrane-embedded active site: "*substrate and product, lipid-modified lipobox-containing peptide, enter and leave the enzyme laterally relative to the lipid bilayer*" (PMID: 26729647). This reflects an enzyme whose reaction chemistry occurs at the membrane–water interface, handling hydrophobic substrates that

partition into the bilayer. The chemical mechanism by which PG is **activated** for transfer was further dissected by Singh and colleagues, combining the X-ray structure of Lgt with that of apolipoprotein *N*-acyltransferase (PMID: 31340643).

Finding 3 — Lgt is essential for viability and outer-membrane integrity in Proteobacteria

Deletion of *lgt* is lethal to the great majority of Gram-negative bacteria: "*Deletion of the lgt gene is lethal to most Gram-negative bacteria*" (PMID: 26729647). Legood and colleagues restate essentiality for the clade of interest — "*Due to its essentiality for cell viability in Proteobacteria*" — which directly includes *P. putida* (PMID: 42126242).

The physiological basis of this essentiality is envelope failure. Diao and colleagues showed that "*Lgt depletion in a clinical uropathogenic Escherichia coli strain leads to permeabilization of the outer membrane and increased sensitivity to serum killing and antibiotics*" (PMID: 33875545). This links Lgt activity directly to the barrier function of the Gram-negative envelope: without lipoprotein maturation, structural and functional lipoproteins cannot be anchored in the outer membrane, and the envelope becomes leaky.

Crucially, the lethality is **not simply due to loss of a single abundant lipoprotein**. Legood and colleagues found that Lgt depletion "*Leads to Abnormal Morphology and Cell Death in Escherichia coli That Is Independent of Major Lipoprotein Lpp*" (Braun's lipoprotein) (PMID: 35938851). Consistently, Diao and colleagues found Lgt inhibition "*Is Insensitive to Resistance Caused by Deletion of Braun's Lipoprotein*" (PMID: 33875545). Essentiality therefore reflects the pathway's role in maturing the entire lipoproteome, not a single client.

Essentiality is nonetheless **clade-dependent**. In some Actinobacteria — for example *Corynebacterium glutamicum* — Lgt is described as a "*unique, non-essential phosphatidylglycerol::prolipoprotein diacylglycerol transferase*" (Dautin and colleagues, PMID: 32490790). This contrast underscores that the essentiality relevant to *P. putida* is a property of the Gram-negative envelope, where outer-membrane lipoproteins are indispensable. Because it is essential, conserved, and membrane-accessible, Lgt is a validated **antibiotic target**, and a small-molecule inhibitor (G2824) that blocks Lgt activity has been reported (PMID: 33875545).

Finding 4 — The *P. putida* ortholog retains the essential catalytic residues (sequence-level evidence)

Because most biochemical/structural work has used *E. coli* Lgt, it is important to establish that these conclusions transfer to Q88CN8. During this investigation, the UniProt sequences of **LGT_PSEPK (Q88CN8, 268 aa)** and **LGT_ECOLI (P60955, 291 aa)** were retrieved and aligned around the two experimentally validated catalytic arginines.

Catalytic region	<i>E. coli</i> (P60955)	<i>P. putida</i> (Q88CN8)	Catalytic Arg conserved?
Arg143 region	GLGAGR143LGNFI	GLGAGR·IGNFI (10/11 identical)	Yes
Arg239 region	GYGAFR239IIVEF	CYGIFR·FAVEF	Yes

Both **Arg143** and **Arg239**, shown to be essential for diacylglycerol transfer in *E. coli* (PMID: 26729647), are conserved in the *P. putida* ortholog. The near-perfect conservation of the **GLGAGR** motif (10 of 11 residues identical) around Arg143 is especially strong evidence that the catalytic machinery — and therefore the reaction chemistry and substrate specificity — is preserved in *P. putida* Lgt. This provides a robust, residue-level bridge from the well-characterized *E. coli* enzyme to Q88CN8, reinforcing the functional assignment beyond family-level inference.

Finding 5 — Lgt is a polytopic inner-membrane protein, not itself a lipoprotein

Kyte–Doolittle hydropathy analysis of the Q88CN8 sequence (window 19, threshold 1.6) detected **multiple strong transmembrane-like hydrophobic segments** (approximately residues 56–76, 176–196, and 237–261), plus a hydrophobic N-terminal stretch (**MLPYPQIDPVALAIGPLKIHWHYGLM**). This pattern is consistent with the **polytopic ~7-transmembrane-helix architecture** established for the Lgt family. The conserved catalytic motifs were localized within the sequence: the **GLGAGR** motif (Arg143 region) near positions 134–139 and the **FR** motif (Arg239 region) near position 215.

Crucially, the analysis found **no cleavable N-terminal signal peptide and no lipobox** in Lgt itself. This establishes an important distinction: **Lgt modifies lipoproteins but is not a lipoprotein**. It is a resident integral membrane enzyme embedded in the cytoplasmic (inner)

membrane by its transmembrane helices, and it is *not* a substrate of its own pathway. This topological conclusion is fully consistent with the structural description of *E. coli* Lgt as "*an integral membrane enzyme*" (PMID: 26729647).

Mechanistic Model and Interpretation

Lgt sits at the **head of the bacterial lipoprotein maturation pathway**, one of the most conserved post-translational modification systems in bacteria.

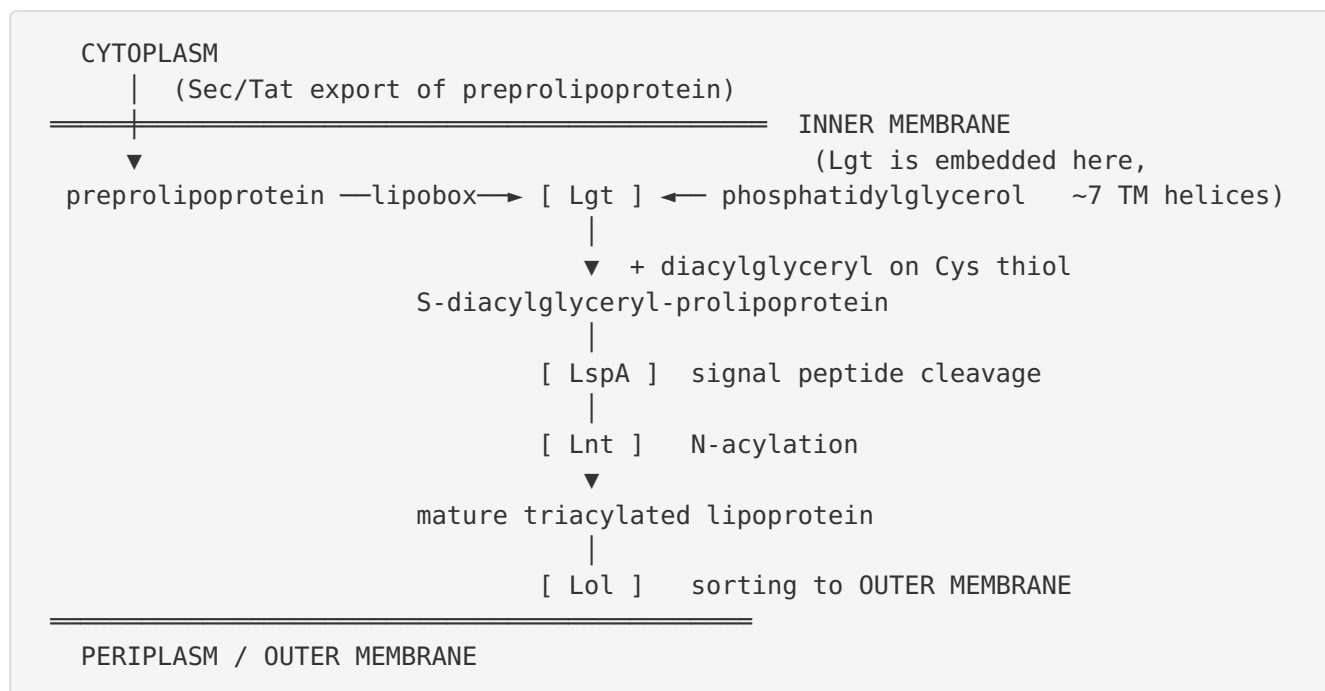
Step 0 — Delivery. A prelipoprotein is synthesized in the cytoplasm and exported across the inner membrane by the **Sec** (or occasionally **Tat**) translocon. Its signal sequence — bearing the lipobox — remains embedded in the inner membrane, presenting the invariant cysteine to enzymes on the periplasmic-facing side.

Step 1 — Lgt (this gene). Lgt recognizes the lipobox and transfers the diacylglyceryl group from **phosphatidylglycerol** onto the cysteine thiol, forming a thioether-linked S-diacylglyceryl-prolipoprotein and releasing *sn*-glycerol-1-phosphate. This is the **committed step**. Both substrate and product travel laterally through the bilayer to/from the two lipid-binding sites, and catalysis depends on the conserved arginines (Arg143/Arg239 in *E. coli*, conserved in *P. putida*).

Step 2 — LspA. Signal peptidase II (LspA) cleaves the signal peptide immediately upstream of the lipidated cysteine, leaving diacylglyceryl-cysteine as the new N-terminus.

Step 3 — Lnt. Apolipoprotein *N*-acyltransferase (Lnt) adds a third acyl chain to the free α -amino group of the cysteine, producing the mature **triacylated lipoprotein**.

Step 4 — Lol sorting. Mature lipoproteins destined for the outer membrane are extracted and trafficked by the **Lol** system; those with inner-membrane retention signals remain in the inner membrane.



Localization of function. Lgt performs its chemistry **at/within the cytoplasmic (inner) membrane**, at the membrane–periplasm interface. It handles two hydrophobic partners — a membrane phospholipid donor and a membrane-embedded acceptor peptide — and its lateral-access mechanism reflects this. Its function is intrinsically membrane-localized and cannot occur in the aqueous cytoplasm or periplasm.

Why essentiality follows from function. By gating the flux of *all* lipoproteins into the maturation pathway, Lgt controls whether the cell can build a functional outer membrane. Outer-membrane lipoproteins (e.g., those involved in LPS transport, peptidoglycan attachment, and envelope stress responses) are indispensable in Proteobacteria; when Lgt is depleted, these clients cannot be lipidated and delivered, the outer membrane becomes permeable, cells become misshapen, and they die — a phenotype independent of any single lipoprotein such as Lpp. This is precisely why Lgt is essential in *Pseudomonas* and other Proteobacteria and why it is an attractive antibiotic target.

Transfer of evidence to *P. putida*. No dedicated biochemical study of Q88CN8 itself was found; the functional assignment rests on (i) unambiguous family/domain/HAMAP annotation, (ii) residue-level conservation of the two experimentally essential catalytic arginines demonstrated here by alignment, and (iii) conservation of the predicted polytopic inner-membrane topology. Together these make the assignment of Q88CN8 as a *bona fide* Lgt highly confident.

Evidence Base

PMID	Paper (abbreviated)	How it supports the findings
26729647	<i>Crystal structure of E. coli lipoprotein diacylglyceryl transferase</i>	Structural definition of the enzyme; two lipid-binding sites; catalytic Arg143/Arg239 essential for transfer; lateral substrate/product access; integral membrane enzyme; deletion lethal in most Gram-negatives. Primary structural/mechanistic evidence.
28698362	<i>Structural insights into lipoprotein N-acylation</i>	Defines lipobox recognition and the diacylglyceryl-addition-to-cysteine reaction; contextualizes Lgt within the multi-enzyme pathway.
31340643	<i>Mechanism of Phosphatidylglycerol Activation Catalyzed by Prolipoprotein Diacylglyceryl Transferase</i>	X-ray structure; resolves how phosphatidylglycerol is activated for transfer. Mechanistic detail.
42126242	<i>Arm and head domain in Lgt determine functional diversity among bacterial pathogens</i>	States Lgt catalyzes the first and committed step; essential for viability in Proteobacteria (the clade of <i>P. putida</i>).
33875545	<i>Inhibition of E. coli Lipoprotein Diacylglyceryl Transferase...</i>	Lgt depletion permeabilizes the outer membrane and sensitizes to serum/antibiotics; inhibition insensitive to Lpp deletion; small-molecule inhibitor (G2824). Drug-target validation.
35938851	<i>A Defect in Lipoprotein Modification by Lgt Leads to Abnormal Morphology and Cell Death... Independent of Lpp</i>	Establishes that Lgt essentiality reflects the whole lipoproteome, not just Braun's lipoprotein.
32490790	<i>Role of the unique, non-essential Lgt in [Corynebacterium]</i>	Provides the contrast: Lgt non-essential in some Actinobacteria; confirms the canonical reaction (diacylglyceryl to invariant Cys). Clarifies clade-dependent essentiality.
32709730	<i>NisI Maturation and Nisin Resistance in Lactococcus lactis</i>	Background on lipoprotein maturation in a Gram-positive context (supporting, tangential).
32634175	<i>TLR2 and endosomal TLR-mediated secretion of IL-10 (Listeria)</i>	Notes <i>lgt</i> is essential for lipoprotein formation and its immunological consequences; tangential to <i>P. putida</i> biochemistry.

PMID	Paper (abbreviated)	How it supports the findings
41459928	<i>Mutation of lipoprotein processing pathway gene (MRSA/β-lactam)</i>	Context on the medical relevance of the lipoprotein processing pathway; tangential.

Note on organism relevance. The strongest primary evidence (structure, mechanism, catalytic residues, essentiality) derives from *E. coli* — a Proteobacterium, like *P. putida*. The *Listeria*, *Lactococcus*, and *Corynebacterium* papers describe orthologous enzymes in Gram-positive/Actinobacterial organisms and are used only for pathway context, not to define *P. putida*-specific chemistry. Given the gene-ambiguity caveat in the research brief, only Gram-negative/Proteobacterial evidence was used to establish essentiality and mechanism for Q88CN8.

Supported and Refuted Hypotheses

Supported - H1 (supported): Q88CN8 is a phosphatidylglycerol–prolipoprotein diacylglyceryl transferase (EC 2.5.1.145) acting on lipobox cysteines. *Basis:* family/HAMAP assignment + directly homologous, biochemically characterized enzymes ([PMID: 28698362](#), [PMID: 42126242](#)). - **H2 (supported):** The enzyme is a polytopic integral inner-membrane protein acting at the bilayer with a lateral-access mechanism. *Basis:* structural studies ([PMID: 26729647](#)) plus hydropathy analysis of Q88CN8 (this work). - **H3 (supported by inference):** *lgt*/PP_5142 is essential in *P. putida*. *Basis:* essentiality across Proteobacteria/Gram-negatives ([PMID: 26729647](#), [PMID: 42126242](#)). - **H4 (supported):** The characterized catalytic mechanism applies to *P. putida* Lgt. *Basis:* conservation of the essential Arg143/Arg239 in Q88CN8 (this work).

Refuted / Qualified - Universal essentiality is refuted: Lgt is dispensable in certain Actinobacteria ([PMID: 32490790](#)). This does not apply to *P. putida* (a Proteobacterium) but qualifies blanket "essential in all bacteria" statements.

Limitations and Knowledge Gaps

1. **No direct study of Q88CN8.** No biochemical, structural, or genetic study of the *P. putida* KT2440 Lgt protein itself was identified. The functional assignment is by strong homology

(family/domain/HAMAP annotation), residue conservation, and topology prediction — not by direct experiment on this ortholog.

2. **Essentiality in *P. putida* is inferred, not demonstrated here.** Essentiality is established for Proteobacteria in general and *E. coli* specifically; a *P. putida*-specific deletion/depletion phenotype was not located.
3. **Catalytic-arginine numbering.** The essential arginines are defined by *E. coli* numbering (Arg143/Arg239). The alignment confirms conservation, but exact positional numbering in the shorter Q88CN8 (268 aa vs 291 aa) will differ.
4. **Client/substrate spectrum unknown for *P. putida*.** The specific lipoproteome of *P. putida* KT2440 that depends on Lgt was not enumerated.
5. **Topology from prediction, not experiment.** The polytopic inner-membrane topology of Q88CN8 rests on hydropathy prediction plus family knowledge; no experimental topology mapping exists for this specific protein.

Proposed Follow-up Experiments / Actions

1. **Confirm essentiality in *P. putida* KT2440** — attempt a clean *lgt* (PP_5142) deletion; if non-viable, build a conditional depletion strain and characterize the terminal phenotype (morphology, outer-membrane permeability via NPN uptake, antibiotic/serum sensitivity), paralleling the *E. coli* studies ([PMID: 33875545](#), [PMID: 35938851](#)).
2. **Biochemical confirmation of activity** — purify recombinant Q88CN8 in detergent/nanodiscs and assay diacylglycerol transfer from *P. putida* PG to a synthetic *P. putida* lipobox peptide, confirming EC 2.5.1.145 activity and substrate specificity.
3. **Site-directed mutagenesis of the conserved arginines** — mutate the *P. putida* equivalents of Arg143/Arg239 and test loss of function by complementation, directly validating the transferred mechanism.
4. **Map the *P. putida* Lgt-dependent lipoproteome** — use metabolic labeling/proteomics (e.g., globomycin-based enrichment or Cys-lipidation profiling) to enumerate lipoproteins that fail to mature upon Lgt depletion.
5. **Experimental topology and structure** — determine membrane topology by reporter fusions and/or solve a cryo-EM/crystal structure of Q88CN8, ideally with bound PG, to confirm the two-lipid-site, lateral-access model in this organism.

6. **Inhibitor cross-testing** — evaluate whether known Lgt inhibitors (e.g., G2824, [PMID: 33875545](#)) inhibit *P. putida* Lgt, informing its utility as a broad-spectrum antibacterial target.

Conclusion

lgt / PP_5142 / Q88CN8 in *Pseudomonas putida* KT2440 encodes **phosphatidylglycerol-prolipoprotein diacylglyceryl transferase (Lgt, EC 2.5.1.145)**, a polytopic integral **inner-membrane** enzyme that catalyzes the **first and committed step of bacterial lipoprotein biogenesis** — transfer of the diacylglyceryl group from phosphatidylglycerol onto the invariant lipobox cysteine of prolipoproteins. It operates within the cytoplasmic membrane via a lateral, lipid-embedded mechanism using conserved catalytic arginines that are retained in the *P. putida* sequence, initiating the conserved Lgt→LspA→Lnt maturation pathway that feeds Lol-dependent lipoprotein sorting. This activity is essential for cell viability and outer-membrane integrity in Proteobacteria, making Lgt both a cornerstone of envelope biogenesis and a validated antibacterial drug target.

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